



Kishoreganj University

3rd Year 2nd Semester B.Sc. (Engg.) Final Examination-2024

Department of Computer Science and Engineering

CSE 3207: Machine Learning (3 Credits)

Time: 3 Hours

Full Marks: 70

Figures shown in the right margin indicate full marks.

Answer 05 out of 07 questions.

1. (a) What is the key difference between traditional programming and Machine Learning using a block diagram? 3
 - (b) Explain Reinforcement Learning with a real-world example. How does reward and punishment help learning? 3
 - (c) Explain the meaning of a linear decision boundary and describe the roles of the weight vector and bias term. 3
 - (d) Consider the decision boundary $x_1 - 2x_2 + 3 = 0$. Find the perpendicular distance from the following points and classify them: 5
 - (i) $x_a = (1, 1)^T$
 - (ii) $x_b = (-1, 2)^T$
 - (iii) $x_c = (3, 3)^T$
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2. (a) Define Fisher's Linear Discriminant Analysis (LDA). 2
 - (b) Explain the bias-variance tradeoff. How does it affect model performance? *Simple?* 3
 - (c) A dataset contains 100 samples: 60 positive, 40 negative. 4
A model predicts:
50 correctly as positive, 10 wrongly as positive, 20 correctly as negative, 20 wrongly as negative. *TP FP*
Construct the confusion matrix and compute:
 - i. Accuracy
 - ii. Precision
 - iii. Recall
 - iv. F1-score
 - (d) What are within-class scatter and between-class scatter? Briefly explain their roles in Fisher's criterion. 5
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3. (a) Why is logistic regression called a probabilistic discriminative model? 3
 - (b) Write the sigmoid function and explain why it is suitable for binary classification. 3
 - (c) What is the role of a cost function in simple linear regression? Provide the formula for the Mean Square Error (MSE) cost function and explain what a smaller value indicates. 3
 - (d) Given $w = [2, -1]$, $x = [3, 1]$, and bias $b = 2$, compute $z = w^T x + b$, then compute the class probability using the sigmoid function and classify the point with threshold 0.5. 5

4. (a) Briefly explain how sigmoid function converts a linear score into probability. 3
- (b) Distinguish among training set, validation set, and test set. 3
- (c) Compare and contrast Ridge Regression and Lasso Regression. In what specific scenario would a data scientist prefer using Lasso Regression over Ridge Regression? 4
- (d) How is boosting different from bagging in terms of training process and error handling? 4
5. *Read the following scenario and answer the question accordingly:*

An email filtering company is designing a simple spam detector using a linearly separable training set. Because the company wants a fast online learning algorithm that updates immediately when a message is misclassified, the engineers choose the perceptron algorithm. They also want to understand when the algorithm converges and how the weights change during learning.

Based on this scenario, answer the following:

 - (a) Explain the perceptron learning rule for a misclassified sample. 3
 - (b) What is overfitting? Why does it occur more in Polynomial Regression? 3
 - (c) State the perceptron convergence condition and mention one limitation of the perceptron model. 4
 - (d) Starting from $w = [0,0,0]$ with learning rate 1 and augmented sample $x = [1,1,0]$ having true label -1, compute the updated weight vector after one misclassification update. 4
6. (a) A model shows very low training error but high-test error. 4
 - i. Identify the problem.
 - ii. Suggest at least two solutions using concepts from regression techniques.
- (b) State the formula for the perpendicular distance of a point from a linear decision boundary. 5
- (c) Explain why logistic regression is suitable for this binary classification task. 5
7. (a) A bank is using a decision tree for loan approval, but managers want to know how the model chooses attributes at each step. Explain entropy in the context of decision tree learning. 6
- (b) What is information gain and why does ID3 use it to select the splitting attribute? 4
- (c) If a dataset contains 8 positive and 8 negative samples, compute the entropy of the class label. 4